A Bidirectional Phenotype Renderer: Singly Even Conjugate Decomposition, Recombination and Shutter Phase

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Abstract

This revision specifies a bidirectional phenotype renderer connecting singly even decomposition, conjugate recombination and shutter phase. Arithmetic splitting and massive-phenotype admission are separated. The upstream pseudo-prime working description, internal 2.5-dimensional transition label and structural connections to the initial bulk rendering and black-hole release are made explicit. Version 1.5 adds a conditional two-resolution measurement interface and finite interference, Bell, no-signalling and special-relativistic checks. Earlier Lorentz kinematics are retained and distinguished from the new assumptions.

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Research method: engineering reverse design

Observed structures, constants and phenomena are legitimate verification inputs for WRRA-specific structures, transformations and connections. Revise an interpreter or interface when it fails; count reproduction of the tested reality as compatibility and explanatory achievement. Independent predictions are additional achievements, not prerequisites. Identify unmeasured quantities calculated after freezing the rules separately as predictions. Evaluate in the order verified inputs → WRRA transformation → outputs → falsification conditions, with adopted assumptions and executed scope disclosed.

Bidirectional rendering and conjugate transitions

This v1.5 revision preserves the 20 October 2026 standalone v1.4 content and adds the two-resolution measurement interface and finite compatibility audit. Previous 630-case Lorentz inputs and numerical results are unchanged.

135. Alternating Single-Sector Output, Remainder Placement and Conjugate Conversion

Added 20 October 2026 · Structural model and initial-condition hypothesis · No newly calculated microscopic production or annihilation rates

Verified inputs

Corresponding particles and antiparticles have equal masses and spin magnitudes, with conjugate internal quantum numbers. Opposite electric charge alone does not identify the pair: an electron and a positron correspond, whereas an electron and a proton do not. A representative confirmed channel is $e^- + e^+ \to \gamma + \gamma$; allowed channels depend on initial states and conservation laws. The incoming particles need not have been created together [S1–S2].

This hypothesis distinguishes which sector a declared source newly outputs in a particular execution from how already existing conjugate phenotypes interact. Requiring pair production in the source-output specification and allowing subsequent physical pair production/annihilation are distinct contracts.

WRRA transformation

135.1 One newly output sector per declared execution and region Let k be an execution index in a declared region, with a source setting $\kappa_k \in \{+1, -1\}$. The signs select matter and antimatter respectively. Let $J_k \geq 0$ be the attempted new source output and $Q_k \in \{0, 1\}$ the resource, interaction and conservation permission gate. This revision maps k to the existing conditional shutter clock of Chapter 112 as a construction assumption; the physical output unit for J still needs a microscopic bridge.

$$J^{\text{new}}_{M,k} = Q_k \frac{1+\kappa_k}{2} J_k, \qquad J^{\text{new}}_{\bar{M},k} = Q_k \frac{1-\kappa_k}{2} J_k.$$

The specified gate gives

$$J^{\text{new}}_{M,k} J^{\text{new}}_{\bar{M},k} = 0, \qquad J^{\text{new}}_{M,k} + J^{\text{new}}_{\bar{M},k} = Q_k J_k.$$

This specifies single-sector production at a given execution. Exclusivity follows by definition of the gate, not from a completed microscopic derivation. It does not set the product of already accumulated occupations $N_M N_{\bar{M}}$ to zero. The switching law is specified below; permissions and resource costs remain explicit state data.

The scope is the declared source's initial/basic output. It is not a universal prohibition of experimentally confirmed pair production in subsequent interactions. Replacing every production pathway by this gate would require an extension reproducing those actual reactions.

135.2 Initial setting of our universe Wonsik Choi's proposal is recorded as an initial-condition hypothesis:

> The phenotype renderer can select the matter or antimatter sector. The initial source execution of our universe was prepared in the matter-output condition.

Thus $\kappa_0 = +1$ is applied to that initial source. This explicitly chosen preparation corresponds to observed matter predominance; it is not a calculated necessity for the sign. The conjugate setting remains a permitted candidate, without granting our realization a privileged selector. No actual second universe or antimatter identification of a dark sector is inferred.

Matter output does not mean producing only negative electric charge. Ordinary matter contains negative electrons and positive protons. The setting refers to conjugate matter/antimatter configurations, not spatial reflection or the earlier exterior-algebra parity selection alone.

Initial-condition ownership is explicit. One may implement κ_0 as a function of the existing SOURCE preparation η; if independently selected, it counts as an additional input. That interface is not completed here. In the baseline branch retaining Chapter 8.1's No Extra Asymmetry Register, require $\kappa_0 = F(\eta, \text{LAW})$. Allowing an independent sign as a new input is an alternative that relaxes that prohibition; do not impose the premises of both branches simultaneously. Chapter 8's prohibition on arbitrary asymmetry from both CP-symmetric LAW and CP-symmetric preparation is retained. The new $\kappa_0 = +1$ preparation declares an asymmetric state and is a different hypothesis.

Actual one-sector particle production requires source energy, charge and every declared conserved or violated quantum number to be tracked. A closed initial state with zero baryon number and exact baryon conservation cannot acquire net baryon number without an owner. Transfer, permitted violation or the opening interface must be supplied by a microscopic model. Output permission and actual production are distinct responsibilities.

135.3 Alternating shutter phases and rare late output The revised proposal specifies the switching law:

$$\kappa_{k+1} = -\kappa_k, \qquad \kappa_0 = +1, \qquad \kappa_k = (-1)^k.$$

The setting alternates at each shutter timing; actual generation occurs only when Q_k permits it. This is not mandatory production of one particle per shutter cycle. Define J_k as a count of comparable output packages. Assume the large initial output $Q_0 J_0 = A > 0$ completes within the matter-phase shutter opportunity. If the initial burst spans several shutter cycles, its weighted sum $\sum_k \kappa_k Q_k J_k$ must be re-

computed. Equal output in two consecutive modes cancels their initial imbalance. An initial sign alone therefore does not ensure matter dominance.

Late output is small and infrequently permitted. Add the condition that the two modes have equal mean permitted output. For example, let $J_k = 1$ for $k = 1, \ldots, 2n$, with independent $Q_k \sim \text{Bernoulli}(p)$ and $0 < p < 1$. The late count difference $D = N_M^{\text{late}} - N_{\bar{M}}^{\text{late}}$ then satisfies

$$\mathbb{E}[D] = 0, \qquad \text{Var}(D) = 2np(1-p).$$

Each late sector has expected count np. A long-run 1:1 ratio requires mode-unbiased statistics and sufficiently many events; exact equality in a finite realization is not guaranteed. This independent Bernoulli example demonstrates statistical compatibility, not a derived microscopic production probability. Mode-dependent permissions can produce a nonzero mean difference.

The total sector difference is

$$\Delta N = A + \sum_{k=1}^{2n} (-1)^k Q_k J_k.$$

A large initial matter output and mean-symmetric late output coexist under these conditions. Baryon asymmetry requires separate weighting by each package's baryon number. The expression does not derive the observed five-percent matter fraction or baryon-to-photon ratio.

135.4 Equal decomposition and recombination of a singly even number A singly even number is divisible by two but not by four [S5]. For positive integers,

$$r = 4j + 2 = 2n, \qquad n = 2j + 1, \qquad j \geq 0.$$

This channel admits only exact halving of such a parent:

$$\mathcal{D}_{\text{eq}}(r) = \{(n, n) : n \in \mathbb{N} \text{ is odd}, \ r = 2n\}.$$

Compositeness is not part of this arithmetic definition. Both $10 \to 5 + 5$ and $18 \to 9 + 9$ are equal odd decompositions; $24 \to 9 + 15$ and $54 \to 9 + 45$ are forbidden unequal decompositions. Code 24 has an even half and is not a candidate in this channel. Actual massive-phenotype implementation is decided separately by the dimensional filter and permission readout. The primality of 5 does not forbid arithmetic decomposition, but neither does decomposition make 5 a three-dimensional matter phenotype.

Equal particle-antiparticle rest masses motivate assigning the same implementation code n to corresponding conjugates of one species. This is a WRRA mapping assumption, not a proof that equal masses imply identical information addresses.

$$\mathcal{C}_{\text{up}}(2n, \xi) \rightleftarrows (n, a, \chi) + (n, a, \bar{\chi}).$$

Left-to-right is pair-production decomposition and rendering; right-to-left is annihilation recombination and phenotype release. Actual outputs require species, conjugate quantum numbers, energy-momentum and phenotype permission. Equal codes and rest masses do not require equal kinetic energies in every frame. The state ξ owns momenta, spin, phase and correlations not recoverable from the aggregate code alone.

Observed pair production remains a reproduction target [S3]. A single photon alone cannot produce a free massive pair in vacuum; the required second photon, recoiling source or external field must be included. Arithmetic admission does not replace physical conservation.

135.5 Separate source, conversion and occupation ledgers Let $P_{M,k}$ and $P_{\bar{M},k}$ count actual pair-derived outputs on each tick. Replace the former simultaneous $+P_k$ terms by

$$N_{M,k+1} = N_{M,k} + J_{M,k}^{\text{new}} + P_{M,k} - H_k,$$
$$N_{\bar{M},k+1} = N_{\bar{M},k} + J_{\bar{M},k}^{\text{new}} + P_{\bar{M},k} - H_k.$$

A permitted pair reserves its second output for the next tick. Pending occupations $L_{M,k}, L_{\bar{M},k}$ record

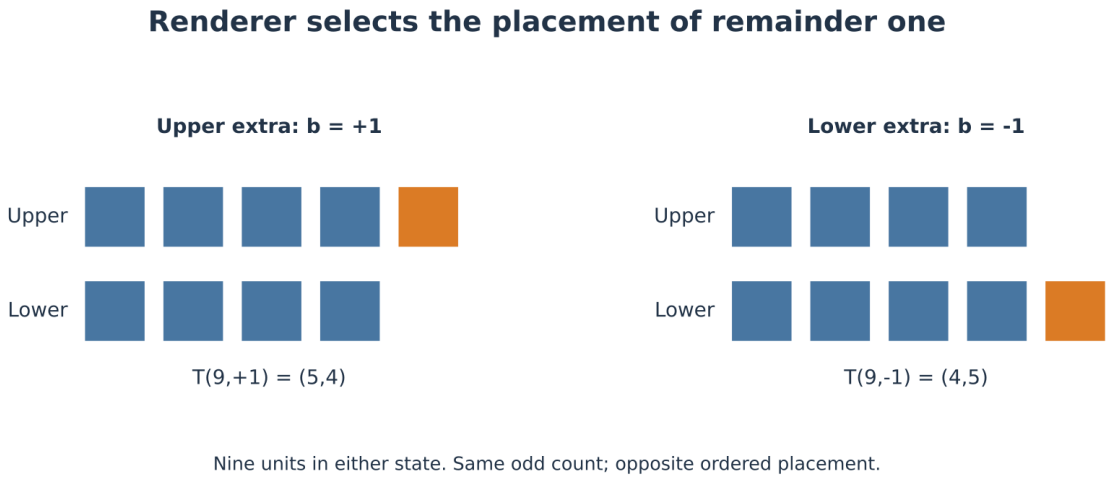
the unrendered partner. The parent/source/environment must own its energy and conserved charges until output; neither gaps in conservation nor double counting are allowed. Over each completed pair, $\sum_k P_{M,k} = \sum_k P_{\bar{M},k}$. Completed symmetric pairs do not create a net matter excess, although the rendered occupations alone can differ during the intermediate tick. Annihilation cannot exceed actual available occupations.

Electric charge is distinct from sector identity. A lone emitted electron requires the opposite charge in the source/environment, with an open ledger until transfer is specified. A neutral matter package is possible, but its baryon and lepton ownership remains required. Average cancellation between modes cannot replace local charge conservation.

135.6 Upper/lower placement of the odd remainder For an odd $n = 2q+1$, introduce $b \in \{+1, -1\}$ to specify whether the remainder one is attached to the upper or lower position. The ordered occupations are

$$T(n,b) = \left(\frac{n+b}{2}, \frac{n-b}{2}\right), \qquad U + D = n, \quad U - D = b.$$

The upper position has one extra unit for $b = +1$; the lower has one extra for $b = -1$. Arithmetic parity is unchanged, but the ordered placement differs. The renderer selects the permitted phenotype's remainder placement b. An interface to the source mode can declare a species reference $b_a^{(0)}$ and assign $b_{a,k} = \kappa_k b_a^{(0)}$. This correspondence is a structural assumption.

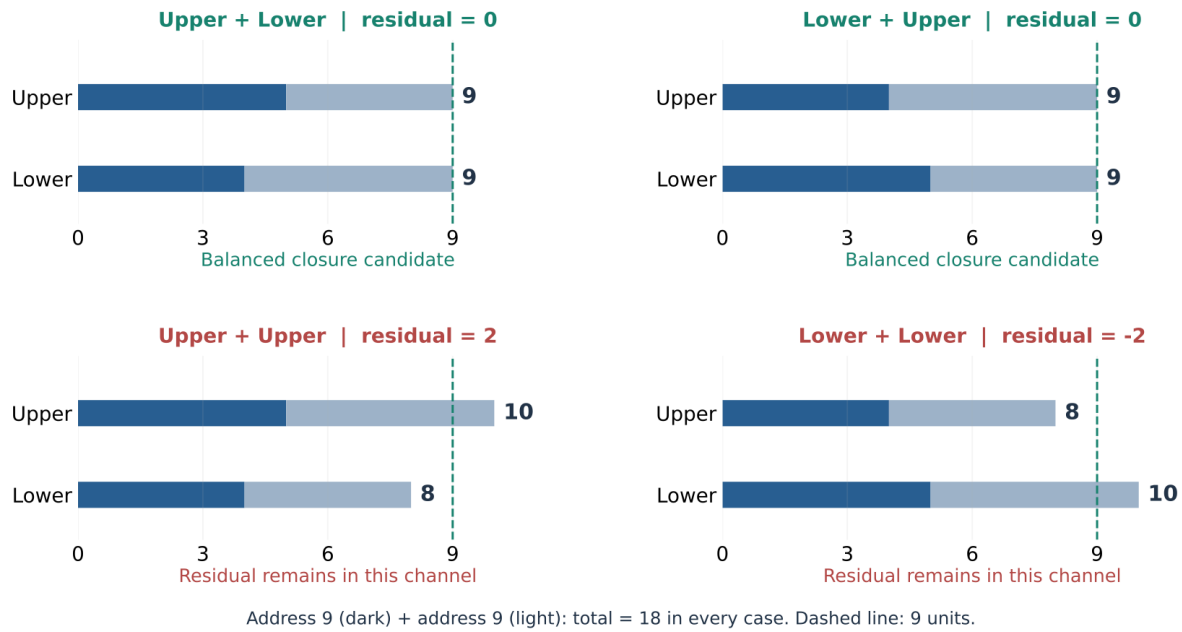


The combined placement residual is

$$\delta_{\text{pair}} = (U_1 + U_2) - (D_1 + D_2) = b_1 + b_2.$$

Opposite placements give zero; like placements leave $+2$ or -2. When $b_2 = -b_1$, each position holds n. For example, the upper placement of 9, $(5,4)$, and lower placement of 9, $(4,5)$, combine as $(9,9)$. Two upper placements instead give $(10,8)$, leaving residual two.

First	Second	Ordered totals for 9 and 9	Residual	This balanced-closure channel
Upper	Lower	$(9,9)$	0	Candidate allowed
Lower	Upper	$(9,9)$	0	Candidate allowed
Upper	Upper	$(10,8)$	$+2$	Closed under this rule
Lower	Lower	$(8,10)$	-2	Closed under this rule



The candidate closure gate is $K_{\text{close}} = \mathbf{1}[n_1 = n_2]\mathbf{1}[b_1 + b_2 = 0]$. The proposed difficulty of interlocking like placements is formalized as non-admission to this particular balanced-closure channel. It does not prohibit all interactions or binding between like placements. Actual interlocking feasibility and transition rates are not calculated from the occupation difference alone.

Exchanging the positions gives $\tau T(n, b) = T(n, -b)$ with $\tau^2 = 1$. This is a candidate geometric representation of conjugacy, not a substitute for the full species and charge/baryon/lepton conjugacy conditions. Upper/lower initially label internal or local positions, not electron spin up/down or an absolute cosmic direction. A physical spatial shape additionally needs coordinate-transformation rules, length/energy functions and an interaction.

For inverse splitting, the candidate parent has $U = D = r/2$ and distributes its occupations to identical odd codes $n = n_1 = n_2 = r/2$ with opposite placements $b, -b$. The code r alone does not specify addresses, directions or quantum correlations; ξ and permission conditions remain necessary. Equal physical masses are imposed through the input-to-code mapping; electric-charge magnitudes, photon channels and annihilation cross sections have not been derived from this occupation model.

135.7 Conjugate structures open each other's release pathway The proposed annihilation interpretation is:

> Two conjugate particle phenotypes can open an interaction pathway that releases their stable implementation structures, transferring their particle-form energy into other carrier phenotypes.

The knot/key language denotes structural compatibility and release. Whether there is an actual geometric or topological twist is unresolved. Such a claim requires defined degrees of freedom, invariants, an energy functional and an interaction operator. Equal mass alone does not establish equal bit count or every information measure. Implementation load and distinguishable states/correlations require separate ledgers.

Use already admitted odd-composite mass-implementation codes n, n, a corresponding species a and conjugate internal structures $\chi, \bar{\chi}$. A candidate code is

$$(n, a, \chi) + (n, a, \bar{\chi}) \xrightarrow{\text{allowed interaction}} (r = 2n, \text{carrier output state}).$$

Every sum of two odd integers is even. Conjugacy therefore gates the interpretation as annihilation, rather

than the arithmetic fact of evenness. Two same-sector particles also have an even address sum and must not annihilate solely for that reason.

The aggregate code r has a different type from an existing even-composite SOURCE resident. Evenness alone identifies neither a photon nor dark matter. Exceeding the existing address cutoff does not automatically create SOURCE capacity; finite output encoding and its cost need specification.

135.8 Energy and information transfer Lost incoming particle occupations must transfer their energy, momentum and conserved internal quantum numbers to the outgoing state. Antimatter energy is not cancelled as negative energy. Zero rest mass of individual outgoing photons does not require zero invariant mass of the whole outgoing system.

$$E_{\text{out}} = E_{\text{in}}, \qquad \mathbf{p}_{\text{out}} = \mathbf{p}_{\text{in}}.$$

Track states, phases and correlations rather than declaring information deleted. The equal-split sum code recovers n but does not preserve momenta, spins, phases or correlations. Asymmetric alternative decompositions are forbidden here. An implementation requiring overall reversibility must retain the distinction in other output degrees of freedom or included environmental correlations. This does not require storing the universe's entire past as an additional archive.

135.9 Existing renderer clock and alternating conjugate placements The conditional shutter already calculated in Chapter 112 is adopted as one renderer tick, without fitting a new time constant:

$$\Delta t_{\text{clk}} = 1.4813019664158633 \times 10^{-21}\ \text{s}, \qquad f_{\text{clk}} \simeq 6.750818015 \times 10^{20}\ \text{Hz}.$$

Matter and antimatter denote conjugate implementations of the particle itself: opposite twist directions or upper/lower placements of remainder one are candidate representations of that distinction. Molecular chirality, amino-acid L/D forms and sugar handedness are excluded. These are two conjugate alternatives, not four independent states or a four-tick cycle.

$$\kappa_{k+1} = -\kappa_k, \quad b_{a,k} = \kappa_k b_a^{(0)}, \quad \tau_{k+1} - \tau_k = \Delta\tau_{\text{clk}}.$$

Choosing which placement labels matter is a reference-sign convention. The same sector returns after $2\Delta t_{\text{clk}}$, corresponding to $3.375409007 \times 10^{20}$ Hz. Clock activity does not require particle output on every tick: permission is still required. Once a pair is permitted, the second output is reserved for the next tick and its intermediate conservation ledger must close. Adopting this existing conditional clock is a construction assumption, not a measurement of a unique universal minimum time.

135.10 One-tick sequential output and initial separation The two equal-code conjugate phenotypes of a pair are stipulated to appear one tick apart. The earlier phenotype can move before its partner appears. Their separation immediately after the second output is

$$d(t_{k+1}) = \left| \mathbf{x}_1(t_k) + \int_{t_k}^{t_{k+1}} \mathbf{v}_1(t)dt - \mathbf{x}_2(t_{k+1}) \right|.$$

For a fixed common source location and constant first-particle velocity, $d = |v_1|\Delta t_{\text{clk}}$. The light-speed reference scale in that source frame is

$$c\Delta t_{\text{clk}} = 4.440831575520451 \times 10^{-13}\ \text{m} = 0.4440831575520451\ \text{pm}.$$

This is a reference length, not a universal actual displacement of a massive particle. Sequential output permits initial spatial separation; subsequent trajectories and interactions require their own dynamics. If timing correlations disagree with observations, or if the parent/environment cannot close intermediate conserved quantities, this sequential rule fails. Production rates, microscopic correlations and cross sections have not yet been calculated.

135.11 Lorentz consistency of sequential output and the existing light-transport framework Verified inputs. We retain the 15-channel, 64-shutter transport check in Chapter 112 (maximum relative speed

discrepancy approximately 6.24×10^{-14}), the seven-direction, 64-shutter check in Chapter 117 (approximately 2.80×10^{-13}), the existing clock interval and its local-frame interpretation. Those checks reproduce the adopted light speed and directional consistency. The present calculation connects this framework to the two output events and intermediate conservation ledger of Chapter 135. Relativistic proper time and four-vector transformations provide verification and reinforcement [S4].

WRRA-specific transformation. Existing clock → local time on an effective source worldline → reservation of conjugate output → intermediate-state conservation → next output → frame-dependent readout. Physical proper time is not assigned to prephenotypic SOURCE addresses. Here, the source is an effective physical source/environment that owns the output events and energy-momentum in spacetime.

(a) Making the local clock explicit. The clock in section 135.9 is defined in the source rest frame. For this sequential-output interface, we adopt that interval as the proper-time interval along the source worldline. This construction makes the earlier local-time definition specific to sequential output.

$$\Delta\tau_{\text{clk}} = 1.4813019664158633 \times 10^{-21}\ \text{s}, \qquad \tau_{k+1} - \tau_k = \Delta\tau_{\text{clk}}, \qquad \kappa_{k+1} = -\kappa_k.$$

Each source's k is its own worldline execution index. Equal indices at distant sources do not impose absolute simultaneity. Any required phase relationship between sources must be specified through physical preparation and interaction. The conjugate label κ and code n are not coordinate-axis names: changing inertial frame does not change matter into antimatter or alter $2n \to n + n$. Upper/lower placement denotes local internal structure, without a universal fixed vertical axis. Complete spin and internal-quantum-number dynamics remain a separate specification.

(b) Transforming the output events. Use metric signature $(+,-,-,-)$. An effective source moving inertially between outputs has four-velocity u^μ satisfying $u_\mu u^\mu = c^2$. The event displacement is

$$\Delta z^\mu = u^\mu \Delta\tau_{\text{clk}}, \qquad \Delta z_\mu \Delta z^\mu = c^2 \Delta\tau_{\text{clk}}^2 > 0.$$

In its rest frame, $\Delta\mathbf{x} = 0$ and $\Delta t = \Delta\tau_{\text{clk}}$. In a frame moving at velocity V relative to that frame,

$$\Delta t' = \gamma_V \Delta\tau_{\text{clk}}, \qquad \Delta x' = -\gamma_V V \Delta\tau_{\text{clk}}, \qquad \gamma_V = (1 - V^2/c^2)^{-1/2}.$$

Thus $\Delta t' > 0$ under time-orientation-preserving Lorentz transformations. The first output precedes the second in every such frame. Coordinate-time intervals change; the proper-time interval remains one tick. An accelerated source requires $\Delta z^\mu = \int u^\mu d\tau$, rather than the constant-four-velocity expression. Event ordering along a future-directed timelike worldline also remains unchanged in that case.

Relative speed V/c	γ_V	Coordinate-time interval between outputs (s)
0	1	$1.4813019664 \times 10^{-21}$
0.6	1.25	$1.8516274580 \times 10^{-21}$
0.8	1.666666667	$2.4688366107 \times 10^{-21}$
0.99	7.088812050	$1.0500671229 \times 10^{-20}$

The source execution rate per coordinate time in this example is $f' = f_0/\gamma_V$. This is distinct from the Doppler frequency of received light signals. The existing shutter is not interpreted as a photon's proper time.

(c) Executing the intermediate ledger with recoil. Use $c = m = 1$ for the calculation. Assign the equal-rest-mass conjugate pair four-momenta $p_1 = (E, p, 0, 0)$ and $p_2 = (E, -p, 0, 0)$, with $E = \sqrt{1 + p^2}$. Choose an initially neutral source $P = (10, 0, 0, 0)$ and a final neutral remnant $q = (10 - 2E, 0, 0, 0)$. These are consistency-test inputs, not fitted masses of new particles.

$$P = p_1 + R, \qquad R = p_2 + q, \qquad R = P - p_1.$$

After the first output, R owns the opposite charge and remaining energy-momentum. The intermediate source has $m_R = \sqrt{R_\mu R^\mu} > 0$, recoils with $u_R = R/m_R$ in these units, and emits the partner after

one proper-time tick. This u_R determines the output-event displacement. Charge is conserved at each stage: $0 = (+1) + (-1)$ and $(-1) = (-1) + 0$. Applying the same Lorentz transformation to all four-momenta preserves both conservation equations. This is an effective kinematic example with an explicit intermediate state; it does not calculate actual electron-positron production amplitudes or rates.

(d) Reading separation. The expression $d = |v_1|\Delta\tau$ in section 135.10 applies to the restricted nonrecoiling-source rest-frame example. It must not be applied indiscriminately using coordinate time in another frame. For simultaneous separation in the recoiling-source frame, select the event x_1 on the first particle's trajectory satisfying $u_R \cdot (x_1 - z_B) = 0$ and read

$$d_{\text{source}} = \sqrt{-(x_1 - z_B)_\mu (x_1 - z_B)^\mu}.$$

This scalar distance is unchanged when the same source-simultaneous hypersurface is transformed. Different observers can choose different simultaneous hypersurfaces and obtain different distances without contradiction. The existing $c\Delta\tau = 0.4440831575520451$ pm remains the light-speed reference distance in the source rest frame.

Outputs. With fixed random seed 135112117, we executed 630 combinations of 18 directions, seven speed magnitudes from zero through $0.99c$, and five momentum settings. The 126 transformation settings include directional duplicates at zero speed. Checks jointly cover the proper-time interval, conjugate mass shells, conservation at each output, inverse transformation, null transport and separation on the source-simultaneous hypersurface.

Check	Maximum error
Normalized timelike event interval	4.27×10^{-14}
Normalized mass-shell residual	6.84×10^{-13}
Two-stage four-momentum conservation, divided by initial energy 10	2.14×10^{-15}
Normalized inverse-transformation residual of event displacement	2.13×10^{-14}
Relative light-speed discrepancy after transformation	3.42×10^{-14}
Normalized source-simultaneous separation residual	2.71×10^{-14}

The second event remained future-directed in every case. Errors are double-precision numerical residuals. Mass-shell, inverse-transformation and separation residuals use $\max(1, |\text{reference value}|)$ in the denominator. As a negative control, transforming the spatial displacement while forcing every frame's coordinate interval to remain one tick gives a normalized interval of -0.777777778 instead of 1 at $V = 0.8c$. This incorrect absolute-coordinate-time rule does not even preserve a timelike interval and is excluded.

Falsification conditions. This implementation fails if rerunning the fixed inputs produces a normalized residual above 10^{-10}, if a future-directed transformation reverses event order, if the intermediate source requires negative energy or a nonphysical state, or if charge or four-momentum fails to close at either stage. This numerical threshold is an implementation criterion, not an experimental uncertainty. Once connected to an actual channel, disagreement with measured production timing, correlations or conserved quantities requires revising the physical interface.

Conclusion. Retaining the existing light-transport framework, we executed equal-only conjugate splitting, one-clock sequential output, recoil and intermediate bookkeeping as locally covariant kinematics. The clock reuses the existing WRRA input without fitting another time constant. This is a concrete connection achieved for Chapter 135. It does not establish covariance of the complete finite-grid operator under all boosts, interacting amplitudes, microscopic causality or phase synchronization across all sources.

[S4] Richard Fitzpatrick, University of Texas at Austin, Proper time and the linked four-velocity lecture. This source supports the proper-time transformation; it is not observational evidence for the WRRA sequential-

output hypothesis.

135.12 Two information interfaces: topmost upstream, intermediate layer and phenotype Verified inputs and construction premise. Retain the dimensional filter, conjugate relation, equal splitting and conservation ledgers. Define pseudo-primes as the information-exchange medium between the topmost upstream and intermediate layers, and singly even states as the medium between the intermediate layer and phenotype. Do not conflate the two interfaces. Here, pseudo-prime is a working term for WRRA upstream constituents, not an identification with the standard number-theoretic set passing a particular Fermat test. Its exact mapping and generator remain to be specified. The upstream region is therefore not redefined as containing only composites.

WRRA-specific transformation. Separate the renderer's operating domain from the admission domain of massive phenotypes. The renderer acts on upstream structure and is not restricted to odd-composite inputs. Existing prime/even-composite/odd-composite classification reads natural-number codes. Define $\mathcal{A}$ as the internal intermediate-layer mapping from pseudo-prime exchange states to singly even transition codes. Its actual generator and ownership of conserved quantities require specification.

Information interface	Medium	Scope
Topmost upstream ↔ intermediate layer	Pseudo-primes	Each individual exchange event between the two layers
Intermediate layer ↔ phenotype	Singly even states	Conjugate decomposition/ rendering and recombination/ release

An individual topmost/intermediate exchange is not automatically assigned the phenotypic shutter period or a global physical time.

$$S_{\text{top}} \underset{\text{pseudo-primes}}{\overset{\rightarrow}{\leftarrow}} S_{\text{mid}}, \qquad S_{\text{mid}} \xrightarrow{\mathcal{A}} \mathcal{C}_{\text{mid}}(r,\xi),$$

$$\mathcal{C}_{\text{mid}}(2n,\xi) \underset{\text{recombine/release}}{\overset{\text{decompose/render}}{\rightleftarrows}} \Phi_{n,+} + \Phi_{n,-}.$$

The symbols Φ denote phenotypes that actually pass admission. Arithmetic decomposition with a closed phenotype filter leaves an upstream or pending state rather than producing a particle. Production renders the conjugate selected by the current shutter phase first, then its partner after one shutter cycle. The reverse direction is an admitted interaction of existing conjugates; no new assumption forces annihilation to finish in one cycle. Bidirectionality does not establish equal forward/reverse rates or detailed balance.

Explanatory output. Pair production and annihilation exchange information across the intermediate/phenotype interface. Pair production is the appearance of embodied forms through singly even decomposition. Annihilation is their conjugate recombination into a singly even state, releasing the two massive particle phenotypes. What disappears is their particle occupation, not energy or information. Include source/environment and carrier outputs in energy, momentum, charge and correlation accounts. The aggregate code is not automatically a photon; individually massless photons can have nonzero total-system invariant mass.

Falsification conditions. Failure of the upstream-to-code map, admission filter, pending ownership or conservation invalidates the implementation. Arithmetic addition alone establishes neither physical annihilation nor full information recovery.

135.13 Singly even numbers and the internal term "2.5-dimensional transition number" Verified inputs. The parity-based sector readout labels primes as one-dimensional, even composites as two-dimensional and odd composites as three-dimensional. This connects to Chapter 75 and differs from Chapter 130's prime/prime-power/mixed-prime folding classes; Chapter 133 retains both rules separately. For example, 9 has parity label 3 but prime-power folding class 2, whereas 6 has parity label 2 but folding

class 3. Do not use these labels as the same dimension variable. A singly even number $r = 4j + 2$ divides exactly into equal odd halves [S5].

WRRA-specific transformation. Distinguish arithmetic classification d_{class} from the transition-role name d_{trans}. Name the conjugate decomposition/recombination role of singly even states a "2.5-dimensional transition" within WRRA, without overwriting the original classification.

$$d_{\text{trans}}(r) = \text{'2.5-dimensional transition'}, \qquad r \equiv 2 \pmod 4.$$

Definition produced. The label 2.5 names the interface role between a two-dimensional aggregate and three-dimensional phenotype; it is not a computed spatial, fractal or spectral dimension. Not every singly even parent has odd-composite halves. Arithmetic allows $10 = 5 + 5$, but 5 fails the existing three-dimensional matter filter. Code $18 = 9 + 9$ is a phenotype candidate, subject to all remaining permissions. Arithmetic also includes $2 = 1 + 1$: 2 remains prime and 1 retains its separate reference-address treatment. Therefore not all singly even numbers are two-dimensional under the original classification.

Falsification conditions. The label must not alter primality/compositeness or render an unfiltered prime as a massive three-dimensional phenotype. Naming the transition does not establish an extra physical dimension.

135.14 A submodel of Big Bang theory: initial implementation and output phase Verified inputs and construction assumption. Retain expansion, cooling, radiation domination, nucleosynthesis and recombination in existing Big Bang theory, together with MCC 3.0's opening and energy-transfer path, as the higher-level description. Treat the phenotype renderer as a submodel seeking to explain initial matter implementation and matter/antimatter output phase within that history. Bulk rendering denotes the common initial implementation boundary; it does not delete the staged thermal history or provide a replacement cosmology with all present content completed at once. The actual dynamical interface with the higher-level description still requires verification.

WRRA-specific transformation. Let the initial boundary phase be $\kappa_{\text{B}} \in \{+1, -1\}$ and its permitted bulk output $A_{\text{B}} > 0$:

$$N_M^{\text{B}} = \frac{1 + \kappa_{\text{B}}}{2} A_{\text{B}}, \qquad N_{\bar{M}}^{\text{B}} = \frac{1 - \kappa_{\text{B}}}{2} A_{\text{B}}.$$

The matter phase initially renders only the matter sector; the antimatter phase would render only its conjugate sector. Adopt $\kappa_{\text{B}} = +1$ for our universe, without requiring another universe. Preparation ownership and independent-input admission follow the branch distinction in section 135.2. Bulk rendering concerns initial sector content, not the instantaneous completion of today's composites, stars or life.

Explanatory output. Distinguish the initial bulk event from ordinary pair production. The opposite of the entire initial sector is not reserved for the next shutter: partner reservation applies to admitted ordinary pair-production channels. Alternating phase does not repeat the output magnitude, and there is no requirement to supply A_{B} again at the next opportunity. A common boundary phase is a declared initial preparation, not absolute coordinate-time simultaneity in every inertial frame. If boundary implementation spans several local shutter cycles, preparation consistency and the sector-weighted total must be calculated.

Falsification conditions. Revise the initial-condition interface if ownership of initial sector preparation, resources, charge, baryon/lepton numbers or consistency with subsequent thermal history and composition cannot be supplied. Choosing a sign alone does not calculate the observed baryon-asymmetry magnitude.

135.15 A common structure for black holes and ordinary conversion Verified inputs and construction assumption. Use bidirectional conversion and phenotype release. The proposal that renderer output interfaces meet in a black hole is an additional structural assumption. Meeting denotes an output-channel junction or boundary state, not an observed contact surface of a physical machine.

WRRA-specific transformation. In ordinary conditions, pair production and annihilation are individual information-conversion pathways between the intermediate layer and phenotype. The topmost/intermediate connection has a separate pseudo-prime exchange interface. Interpret the black hole as a special output-boundary state in which embodied forms are released where output interfaces meet. The architec-

ture thus places initial bulk rendering, ordinary local decomposition/recombination and black-hole boundary release as three operating regimes of the same renderer.

Explanatory output and scope. Distinguish release of form from information deletion. Energy, momentum, charge, angular momentum and correlations require owners after release. This assumption does not remove black-hole gravitational mass or identify upstream conversion with observable communication through an event horizon. No horizon, metric, entropy, radiation or conversion rate has been calculated from this new interface.

Falsification conditions. Revise the identification if ordinary reactions and the black-hole boundary cannot use consistent conservation rules, or if an implemented metric, thermodynamics or observable behavior contradicts black-hole observations. The earlier 630 Lorentz kinematic cases are not verification of this black-hole hypothesis or the complete Big Bang boundary.

135.16 Shutter terminology and reuse of the conditional time interval **Verified inputs.** Retain the existing interval obtained under Chapter 112's energy and phase-resolution conditions; no time constant is refitted.

Term	Definition
Shutter timing	Event τ_k at which phase and output permission are read
Shutter period/cycle	Proper-time interval $\Delta\tau_{\mathrm{sh}}$ between consecutive output opportunities
Shutter frequency	Opportunity rate $f_{\mathrm{sh}} = 1/\Delta\tau_{\mathrm{sh}}$
Output phase	Matter/antimatter mode κ_k at that opportunity

$$\Delta\tau_{\mathrm{sh}} = 1.4813019664158633 \times 10^{-21}\ \mathrm{s}, \qquad f_{\mathrm{sh}} \simeq 6.750818015 \times 10^{20}\ \mathrm{Hz}.$$

WRRA-specific transformation. Replace the informal click with shutter timing or shutter cycle as appropriate. The calculation fixes a conditional interval, not a measured absolute phase or first event time. Alternate the phase at successive opportunities:

$$\tau_k = \tau_0 + k\Delta\tau_{\mathrm{sh}}, \qquad \kappa_{k+1} = -\kappa_k.$$

Outputs. Reuse the existing period in the bidirectional renderer specification. Some shutter opportunities have no particle output. An admitted pair renders its conjugates one cycle apart. The same-phase repetition rate is $f_{\mathrm{sh}}/2$. These values do not determine shutter-open duration or duty cycle. Section 135.11 gives the proper-time/coordinate-time transformation.

Falsification conditions. Revise the conditional period or physical interface if identical energy/phase inputs fail to reproduce the interval or if observation conflicts with the interface. Neither absolute global simultaneity nor an experimentally established universal minimum time is introduced.

Outputs and scope

Version 1.4 adds sections 135.12–16 on the bidirectional renderer, upstream operating domain, 2.5-dimensional transition label, Big Bang/black-hole connection and shutter terminology. These additions specify structure; they do not add black-hole or thermal-history numerical outputs. The earlier 630 Lorentz-case inputs and results are retained.

The new outputs are an alternating-mode/rare-output statistical ledger, conjugate splitting of even aggregate states with a common annihilation ledger, a single-sector gate for initial/basic source production, the matter-prepared initial condition of our universe, the conjugate-release interpretation and the type distinction between arithmetic codes and physical ledgers, now supplemented by equal-only splitting and the existing-clock sequential output contract. No new microscopic production rate, annihilation cross section/lifetime, cosmic thermal history or five-percent composition calculation was executed. Chapter 135 connects Chapters 8 and 18 to a subsequent output-architecture hypothesis without recalibrating existing address, energy, GR or CMB results.

Falsification conditions

An implementation fails if it annihilates same-sector particles merely because their address sum is even, identifies electron/proton as conjugates merely because charges are opposite, loses or duplicates energy/conserved quantities, or discards information it declares conserved. A microscopic interface must be revised if it fails to reproduce known production/annihilation channels or lacks an owner for its initial preparation and resource costs.

Conclusion and genealogy

The central v1.4 proposal distinguishes pseudo-prime exchange at the topmost/intermediate interface from singly even exchange at the intermediate/phenotype interface. Singly even decomposition implements conjugate embodied forms; reverse recombination releases their massive particle occupations. Output phase alternates on the existing conditional shutter period. Initial bulk rendering, ordinary local conversion and black-hole boundary release are proposed as distinct operating regimes of one architecture. The 2.5-dimensional term names the transition role internally. Physical verification of these new structural interfaces is distinct from the arithmetic and Lorentz audits.

The hypothesis distinguishes a large initial matter-mode output, rare later output under alternating modes, and conjugate splitting of even aggregate states with reverse annihilation. The pair-production prohibition is withdrawn. Our universe's $\kappa_0 = +1$ is a declared initial condition; arithmetic evenness is a candidate transition code. The upper/lower sketch illustrates conjugate fit rather than specifying the actual geometry, which may be more complex. Actual knot structure and microscopic interactions remain to be derived.

This chapter is added to the formal v0.24 baseline. The earlier v0.25 calibrated generation/survival draft is discarded; this material replaces it in the new v0.25 without incorporating that calculation. The separated matter/antimatter-domain and SOURCE-size exploration also remains separate. Inclusion here records a structural hypothesis, not completed physical validation.

[S1] MIT, Markus Klute, Charge Conjugation, Lecture 2.4.

[S2] MIT, Principles of Radiation Interactions: Electron–Positron Annihilation.

[S3] SLAC E144 Collaboration, Experiment E144: pair production; K. T. McDonald, Positron Production by Laser Light, 1997.

[S5] Eric W. Weisstein, Singly Even Number, Wolfram MathWorld. This defines the existing arithmetic term; the 2.5-dimensional transition label is a separate definition introduced in this manuscript.

v1.4 arithmetic audit and computational scope

The audit checks 500 singly even parents from 2 through 1998 and 2,000 upper/lower placement combinations. Of these, 332 pass the odd-composite candidate filter and 168 admit arithmetic splitting but fail that filter. Controls distinguish 10→5+5 from 18→9+9 and retain the 2→1+1 boundary. These results check arithmetic admission versus phenotype admission. The previous Lorentz script was rerun and reproduces the 630-case results. No new physical implementation of the Big Bang, black holes or the upstream pseudo-prime map is calculated.

Two-resolution rendering and measurement: conditional compatibility audit

Verified inputs and engineering criterion

WRRA uses engineering reverse design: observed constants and physical structures are legitimate construction inputs. A mismatch is a reason to revise the interpreter or its interface. Reproducing known phenomena counts as compatibility and explanatory achievement. An independent prediction is an additional achievement, not a prerequisite for that status. A quantity calculated for an unmeasured regime after the model inputs and rules have been fixed is identified separately as a prediction. No such new empirical prediction is claimed by the present finite audit.

The adopted shutter proper period remains $\Delta\tau_{\text{sh}} = 1.4813019664158633 \times 10^{-21}$ s. With $c = 299792458$ m/s its transport scale is $L_{\text{sh}} = c\Delta\tau_{\text{sh}} = 4.440831575520451 \times 10^{-13}$ m, approximately 0.444083 pm. This revision adds a conditional internal lower interval for successive registered updates on the same local apparatus worldline: $\Delta\tau_{\text{obs}} \geq t_P$. Its associated local internal length scale is $\ell_P = ct_P$. The Planck floor is an assumption, not a measured minimum duration or resolution.

Using the 2022 CODATA inputs h, G and c gives $t_P = \sqrt{\hbar G/c^5} \simeq 5.39124645 \times 10^{-44}$ s and $\ell_P \simeq 1.61625502 \times 10^{-35}$ m. These scales inherit the uncertainty of G; extra displayed digits in the JSON serve reproducibility. The ratio $\Delta\tau_{\text{sh}}/t_P \simeq 2.74760574 \times 10^{22}$ is not assumed to be an exact integer. No such number of internal updates was simulated. Importing G does not derive gravity from WRRA.

Complex joint density states, unitary evolution, Born probabilities and projective measurement instruments are explicitly adopted as observed-structure interpreter rules. They have not been generated from the integer code by this audit. Their adoption is legitimate reverse engineering; the new contribution tested here is their connection to the two local update schedules.

WRRA transformation and corrected operating rules

A. Periodic representation and microscopic evolution. The shutter specifies periodic phenotype output on a local source worldline. L_{sh} remains its representation scale, but is not a universal Cartesian lattice, a deletion threshold for shorter states, or a period during which all microscopic physics stops. Microscopic phase, joint correlations and permitted interactions persist between shutter opportunities. Macroscopic representation concerns stable apparatus and environment records. It neither copies an arbitrary unknown quantum state nor measures the whole universe on every shutter.

B. Measurement-triggered conditional records. The measurement permission gate Q_{obs} depends on physical apparatus/environment coupling and record formation, not human attention or consciousness. Premeasurement information owns phase and joint correlations; increasing resolution alone does not determine an outcome. For a prepared state ρ and an allowed instrument M_a, use

$$p(a) = \mathsf{Tr}(M_a \rho M_a^\dagger), \qquad \rho_a = \frac{M_a \rho M_a^\dagger}{p(a)}, \qquad \sum_a M_a^\dagger M_a = I.$$

An established record a is definite for that measurement. Incompatible observables need not all become definite. Unobserved environmental correlations remain part of the inclusive description. This is an adopted operational record rule, not a new solution of every interpretation of measurement.

C. Conditional Planck floor. Successive registered updates on one apparatus worldline obey the assumed local proper-time threshold. ℓ_P specifies a conditional internal refinement limit in that apparatus's local rest frame. It does not promise that every detector resolves ℓ_P, that every measurement completes in t_P, or that position and momentum can both be measured arbitrarily precisely. The two schedules are not synchronized across the universe, and no exact integer Planck lattice is imposed. The scales alone supply neither microscopic dynamics nor a computational resource budget.

D. Relativistic connection. Apply time thresholds to local massive apparatus proper time; do not apply them to arbitrary coordinate gaps between different apparatuses or to a photon's zero proper time. Spacelike-separated measurements have no mandatory common execution order. Their allowed local operations must commute in the implemented model, and joint statistics must be order independent. A conditional change in the description after learning a remote outcome is distinct from physically transmitting that outcome. Changing a remote setting may not alter the other party's nonselective outcome statistics. Preserve entangled joint states rather than replacing them with independent local outcome assignments. Tensor-factor commutativity is supplied in this finite model; interacting-field microcausality has not been derived.

E. Production and measurement permissions. Measurement registration does not advance the matter/antimatter shutter phase or create a new particle by observation alone. Equal-only splitting/recombination, next-shutter conjugate reservation, and source ownership of pending charge and four-momentum retain their previous scope. Apparatus and environment must own measurement energy and momentum

exchanges. A degenerate-pointer, energy-preserving Z-QND instrument is implemented as one example. A general apparatus energy/momentum theory is not claimed.

Rejected interpretations and repairs

1. Treating all premeasurement states as independent classical local numbers does not secure Bell correlations. The 16 deterministic local assignments in the audit have CHSH bound 2. An entangled singlet and the adopted joint measurement rule give $2\sqrt{2}$. The repair retains joint correlations without a controllable superluminal signal. This does not exclude every possible deterministic interpretation.
2. Automatically erasing path phase at a shutter destroys interference. The repair makes periodic macroscopic representation distinct from an unperformed which-path measurement. Actual which-path records require their physical coupling.
3. A literal uniform sampling grid of spacing L_{sh} with hard cutoff $k_{\mathrm{max}} = \pi/L_{\mathrm{sh}}$ would cap a represented photon's energy at approximately 1.395957 MeV. This calculated warning is conditional on that extra hard-grid interpretation; it is not a WRRA prediction. The repair permits shorter microscopic states and treats the shutter scale as representation scope.
4. A common simultaneous global update or identical Cartesian minimum grid in every inertial frame conflicts with the ordinary transformation of coordinates and spacelike event order. The repair uses local proper-time thresholds and local rest-frame scales without privileged global simultaneity. This does not prove covariance of an arbitrary minimum-length dynamics.
5. Definite records do not remove all quantum uncertainty. The repair retains incompatible-observable dispersions and does not equate internal refinement with unlimited experimental precision.

Outputs: executed finite checks

Run `python3 audit.py` with Python 3 and NumPy. The fixed seed is 20261020; `results.json` records the input values, counts, numerical errors, negative controls and limits.

Check	Executed output
Interference, 361 phases	$p_0 = (1 + \cos\phi)/2$; maximum error 5.55×10^{-16}; visibility 1
Premature-dephasing control	Visibility approximately 1.11×10^{-16}
Singlet CHSH	2.82842712474619; local deterministic bound 2 over 16 assignments
Nonselective local instruments	1,000 channels over 200 random mixed joint states; remote reduced-state change at most 2.22×10^{-16}
Spacelike tensor-factor order	4,000 unnormalized branch comparisons; difference at most 1.11×10^{-16}
Probability normalization	Maximum error 3.33×10^{-16}
Repeated same-observable record	Maximum error 2.22×10^{-16}
Pauli dispersion bound	No violation of $(1 - \langle X\rangle^2)(1 - \langle Y\rangle^2) \geq \langle Z\rangle^2$ in the tested states
Local proper intervals	101 boosts, 202 interval comparisons; maximum relative error 9.88×10^{-15}
Conserving measurement example	Unitarity and total-energy commutator errors 0 for the declared QND example
Local scheduler	Four inter-shutter registrations leave the alternating shutter schedule unchanged

The tests deliberately detect failed fringe visibility after premature dephasing, failure of independent local assignments to reach the singlet CHSH value, and noninvariance of an absolute spacelike order. Rejection of a $t_P/2$ interval checks enforcement of the assumed threshold, not its physical truth. The quantum

checks are algebraic compatibility tests using adopted rules, not new experiments or a fit to raw experimental Bell data. They do not discriminate a Planck floor from other sufficiently short candidate floors. They are recorded separately from the retained 630-case audit and the book's earlier 288-case reverse-interpreter audit.

Falsification conditions and remaining scope

Revise a connection if it produces negative or unnormalized probabilities; erases interference without a physical record; fails observed correlations; lets a remote setting alter local nonselective statistics; makes spacelike joint statistics depend on coordinate ordering; or violates inclusive apparatus/ environment conservation. An empirical test of the internal time floor additionally requires a physical identification of the same apparatus's successive processing events and a measurable consequence. A chosen numerical scheduling threshold alone is not that test.

No dynamical curved metric, Einstein equation, quantum gravity calculation, general continuous interacting field, or integer-to-complex-amplitude generator is newly executed. Special-relativistic event kinematics and finite commuting instruments do not establish compatibility with every possible gravitational setting. The two-resolution construction passes its stated finite checks after the repairs above. This is a reverse-engineered compatibility result, while the Planck lower scale remains conditional.

Sources and reproducibility

- NIST, 2022 CODATA complete listing: https://physics.nist.gov/cuu/Constants/Table/allascii.txt.
- Hensen et al., experimental Bell-correlation benchmark: https://arxiv.org/abs/1508.05949. The ideal singlet calculation here is separate from those experimental data.
- Fedida, relativistic measurement consistency and its additional assumptions: https://arxiv.org/abs/2506.14693. The present tensor-factor example is not a reimplementation of the complete field-theoretic result.
- Source, figures, code and results: https://github.com/Wonsik-Choi-janefather/wrra-m-0.1/tree/main/submission/bidirectional_phenotype_renderer_v1_5.
- Related integrated book, v0.25-r4: https://zenodo.org/records/23486923. This standalone v1.5 release adds the present result without declaring that the book's released PDF has already been revised.